

Toast Notification System — 2 Min Recap

Mental Model

Button click ↓ Create toast + unique ID ↓ Add to notification[] ↓ Toast renders ↓ Manual close OR timer expires ↓ removeToast(id) ↓ Filter by ID

State

```
notification = [ { toastID, toastMessage, toastType } ]
```

Array → natural choice because toasts are stackable.

Core Operations

showToast(type) → ADD toast

removeToast(id) → REMOVE specific toast

Important React Rule

If new state depends on previous state:

```
setNotification(prev => [...prev, newToast])
```

```
setNotification(prev => prev.filter(item => item.toastID !== id) )
```

REMEMBER:

New state depends on old state ↓ Use functional update → prev

Auto Close

showToast() ↓ setTimeout() ↓ removeToast(toastID)

Timer callback should capture the toast ID.

Don't expect setTimeout to pass your ID automatically.

Manual Close Mistake

```
onClick={removeToast}
```

React passes: event

NOT: toastID

Instead:

```
onClick={() => removeToast(item.toastID)}
```

Component Structure

App |—— Buttons → trigger toast | |—— Toast |—— toast-box |—— toast-item |—— toast-item
 |—— toast-item

Styling

```
className={toast-item ${item.toastType}}
```

```
.toast-box { /* common positioning/layout */ }
```

```
.toast-box.success { } .toast-box.warning { } .toast-box.info { } .toast-box.error { }
```

Positioning

top-left top-right bottom-left bottom-right

Prefer:

position: fixed

because Toast belongs to the viewport, not normal document flow.

Accessibility

WHY?

Toast is usually visual, but a screen-reader user may not see it.

When a new toast appears:

Visual user ↓ sees notification

Screen-reader user ↓ needs notification announced automatically

That's why we use ARIA live-region semantics.

role="alert"

Use on the INDIVIDUAL toast:

Payment successful

Meaning:

"Something important has appeared and should be announced to assistive technology."

role="alert" is appropriate for:

- Errors
- Warnings
- Important status messages
- Success/failure notifications that need immediate attention

aria-live

aria-live tells the screen reader:

"Watch this region for changes and announce new content."

Accepted values:

`aria-live="off"` → Don't announce changes automatically.

`aria-live="polite"` → Announce when the screen reader is idle. → Doesn't interrupt the current announcement.

`aria-live="assertive"` → Announce immediately. → Can interrupt the current announcement.

role="alert" vs aria-live

role="alert" already has assertive live-region semantics.

Therefore for a simple Toast:

Payment successful

is generally enough.

If using `aria-live` explicitly:

Payment successful

Use **polite** when the notification is useful but NOT urgent.

Use **assertive** only when the user needs to know immediately.

Don't blindly add:

`role="alert" + aria-live="assertive"`

just because both are available.

Close Button Accessibility

Native `<button>` is already keyboard accessible.

But "X" alone may not communicate its purpose well to a screen reader.

Use:

`<button type="button" aria-label="Close notification">`

X

Screen reader hears:

"Close notification"

instead of just:

"X"

Interview Rule

Ask:

"Does this content need to be announced to a screen-reader user?"

If YES ↓ Live region semantics

For Toast: role="alert" OR aria-live="polite/assertive"

For urgent notification: → alert / assertive

For non-urgent notification: → polite

Accessibility Mistakes

 Making Toast a